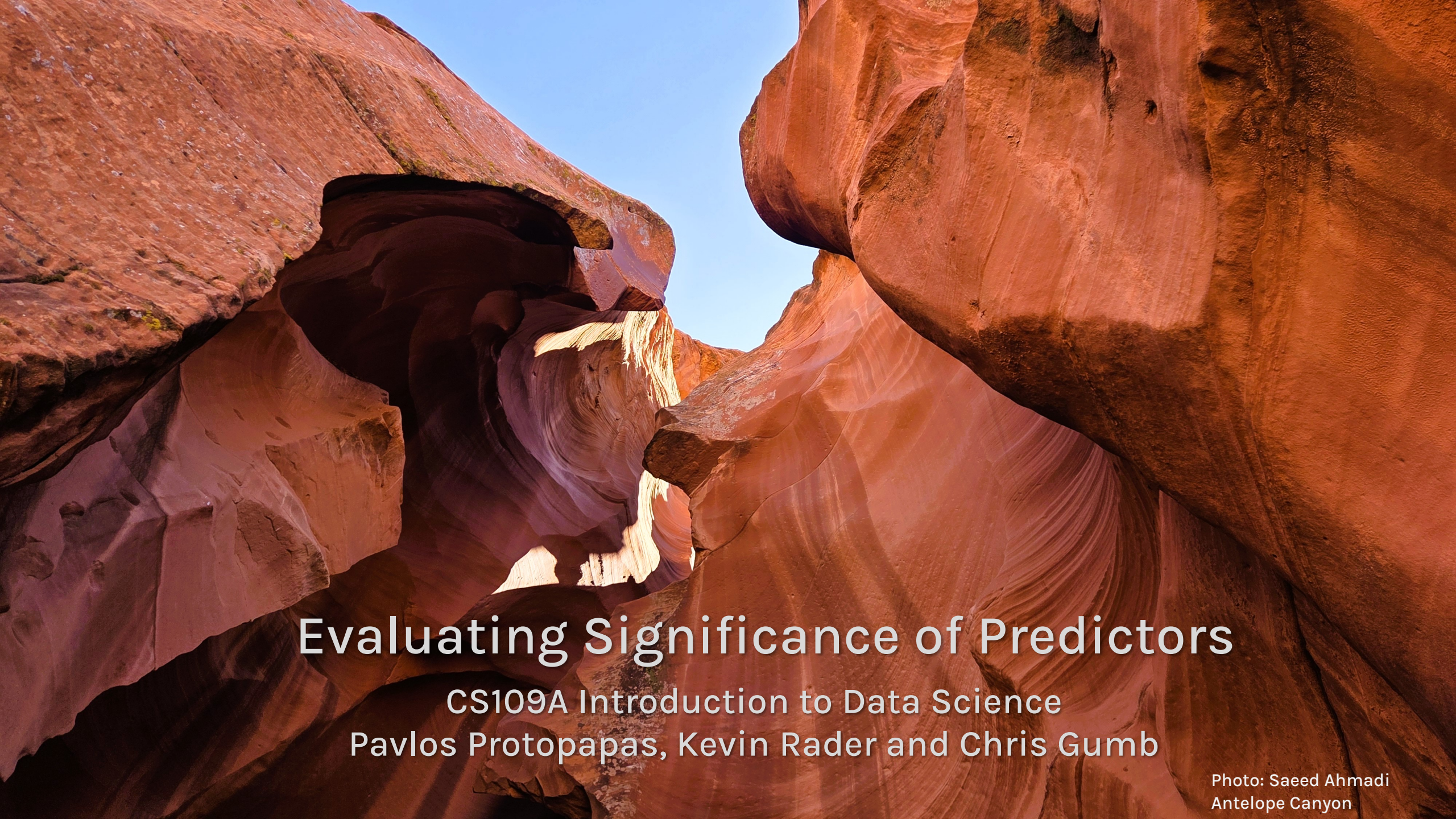


Evaluating Significance of Predictors

CS109A Introduction to Data Science

Pavlos Protopapas, Kevin Rader and Chris Gumb

Photo: Saeed Ahmadi
Antelope Canyon

Outline

Part A and B: Assessing the Accuracy of the Coefficient Estimates

Bootstrapping and confidence intervals

Part C: Evaluating Significance of Predictors

Does the outcome depend on the predictors?

Part D: How well do we know $\hat{f}$

The confidence intervals of $\hat{f}$

How Reliable are the Model Interpretations

Suppose our model for advertising is:

$$y = 1.01x + 0.005$$

where y is the sales in units (each unit is $1000) and x is the TV budget.

Interpretation: For every dollar invested in advertising gets you 1.01 back in sales, which is a 1% net increase.

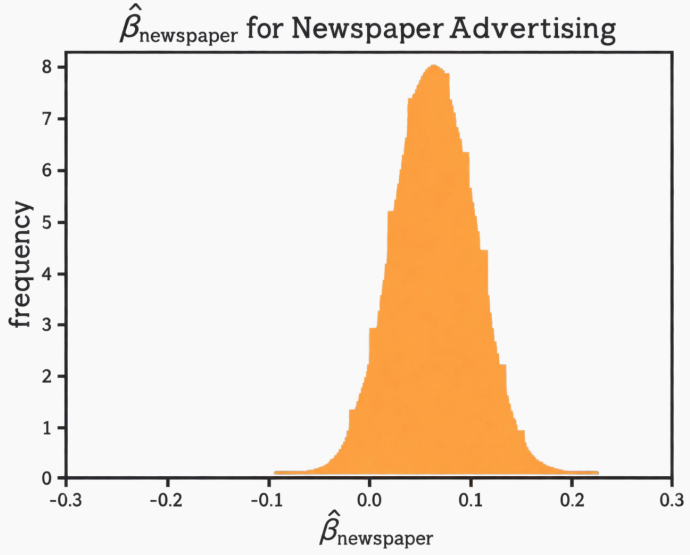
But how certain are we in our estimation of the coefficient 1.01?

Now you know how certain you are in your estimates, will you want to change your answer?

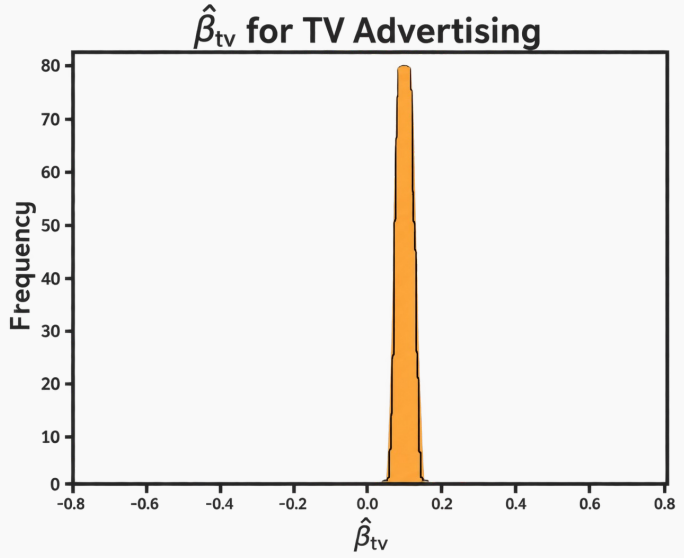
Feature Importance

Now we know how to generate these distributions we are ready to answer ***two important questions:***

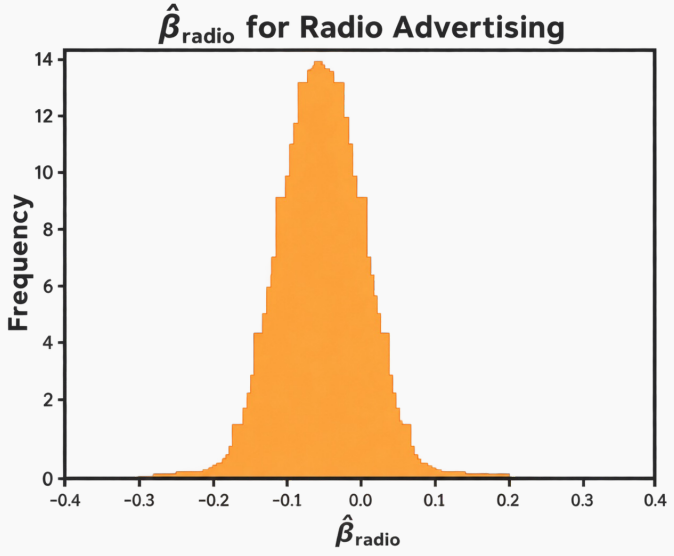
A. Which predictors are most important?
B. And which of them really affects the outcome?



$\mu_{\hat{\beta}_1} = 0.1$
$\sigma_{\hat{\beta}_1} = 0.05$



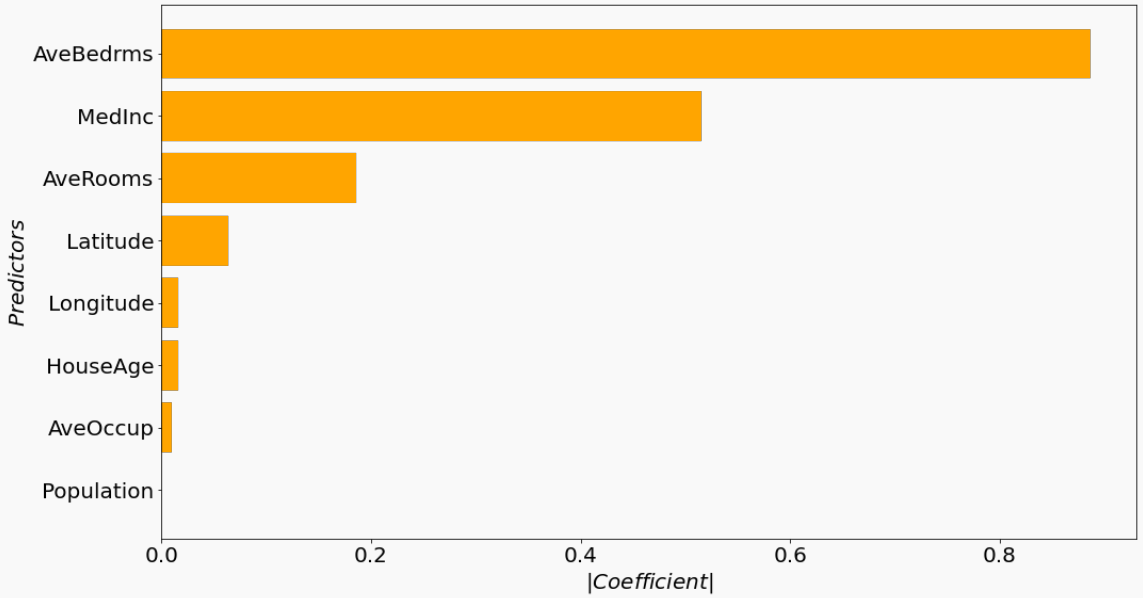
$\mu_{\hat{\beta}_2} = 0.05$
$\sigma_{\hat{\beta}_2} = 0.005$



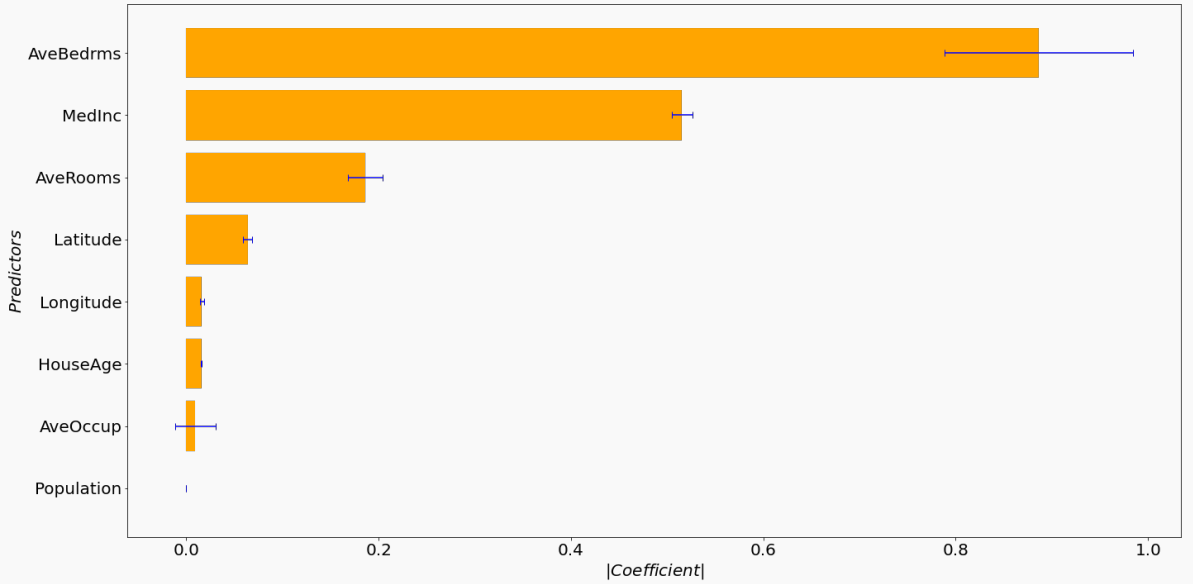
$\mu_{\hat{\beta}_3} = -0.05$
$\sigma_{\hat{\beta}_3} = 0.1$

Feature Importance

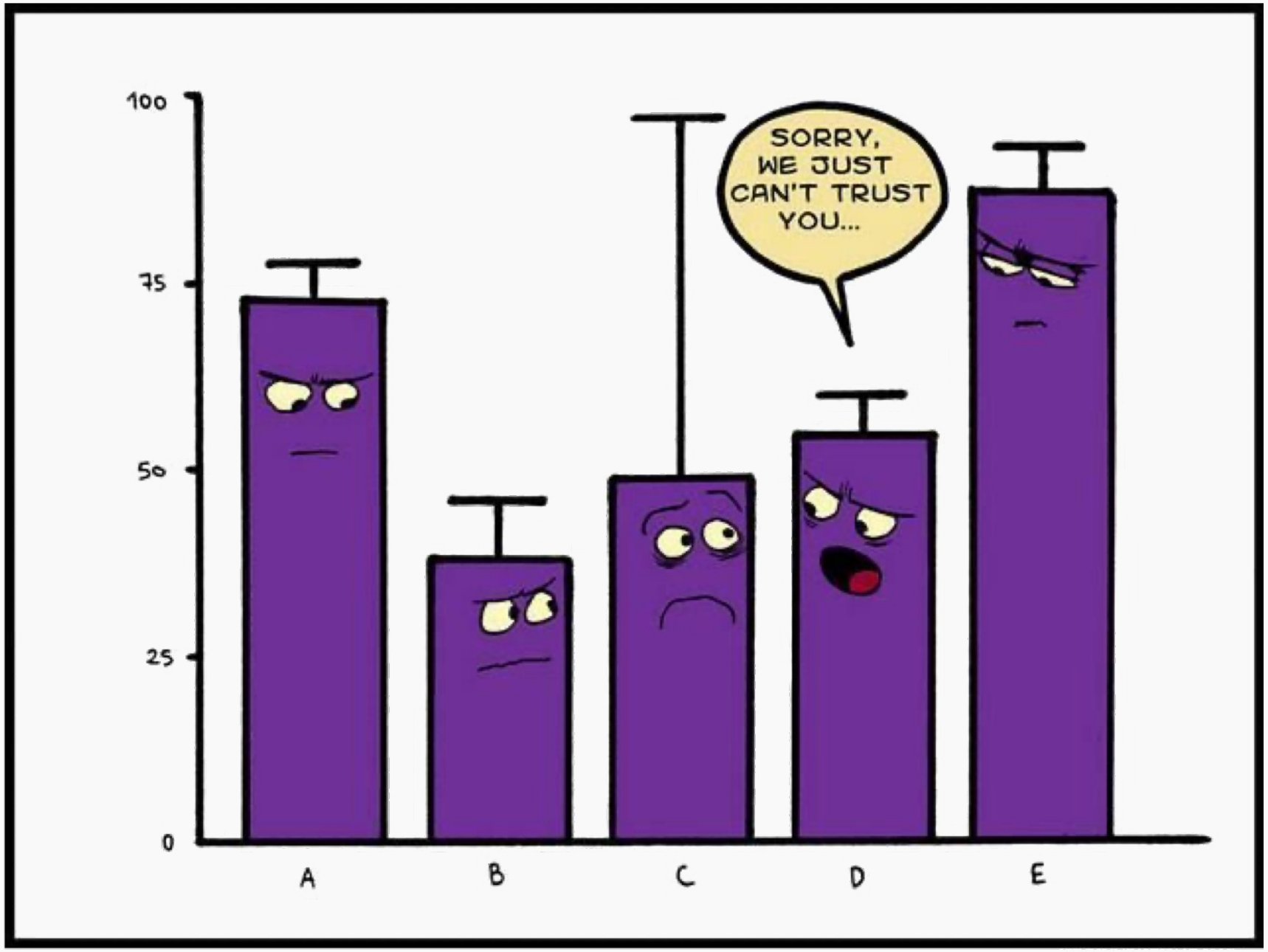
The example below is from California housing data. The coefficients below are from a model that predicts prices given house size, age, crime, etc.



Feature importance based on the absolute value of the coefficients.



Feature importance based on the absolute mean value of the coefficients over multiple bootstraps including uncertainties.


100
75
50
25
0
A
B
C
D
E
SORRY, WE JUST CAN'T TRUST YOU...
WeKnowMemes

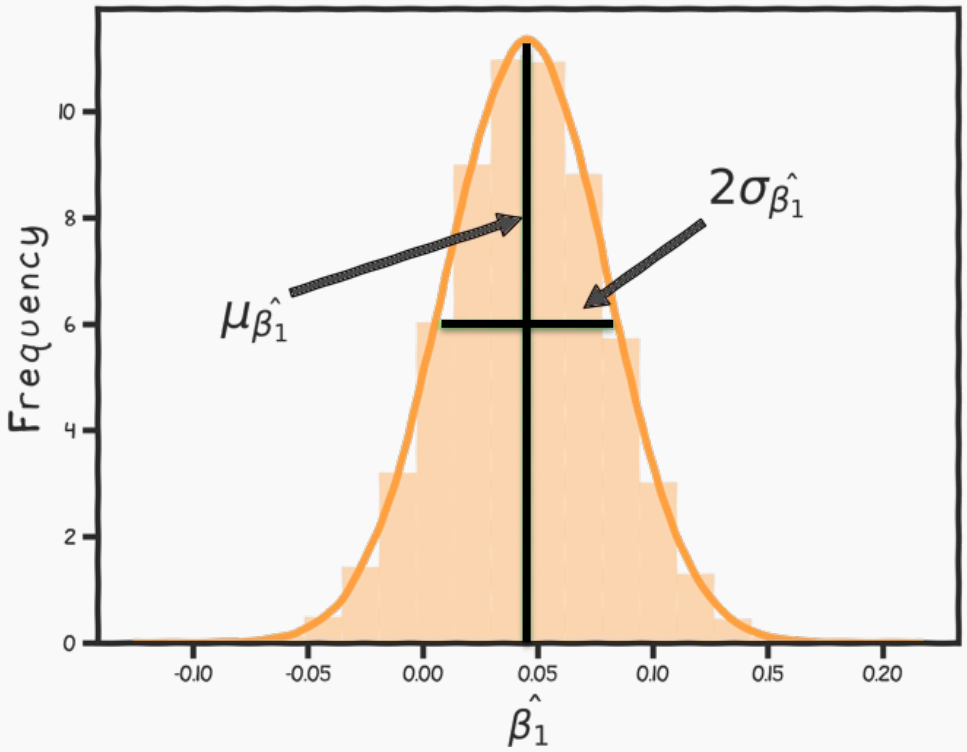
Feature Importance

To incorporate the coefficients' uncertainty, we need to determine whether the estimates of $\beta's$ are sufficiently far from zero.

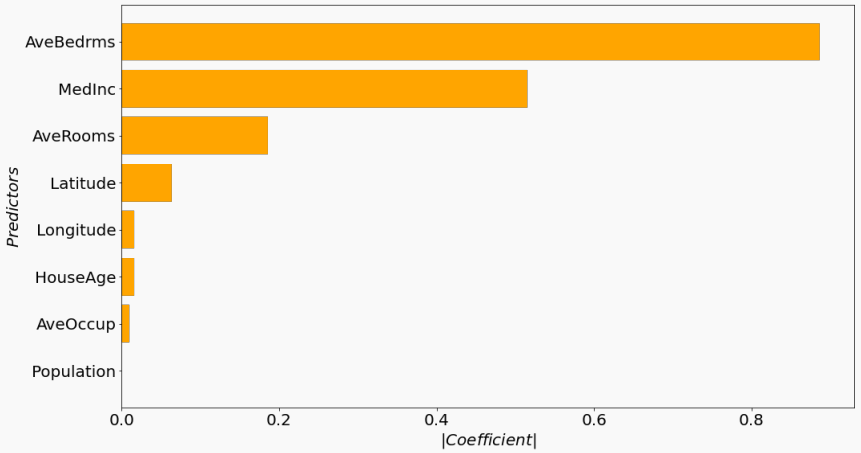
We define a statistic called $\hat{t}$ (bootstrap signal-to-noise ratio), which measures the signed distance from zero in units of estimated standard error.

$$\hat{t} = \frac{\mu_{\hat{\beta}_i}}{\mathrm{SE}_{\hat{\beta}_i}}$$

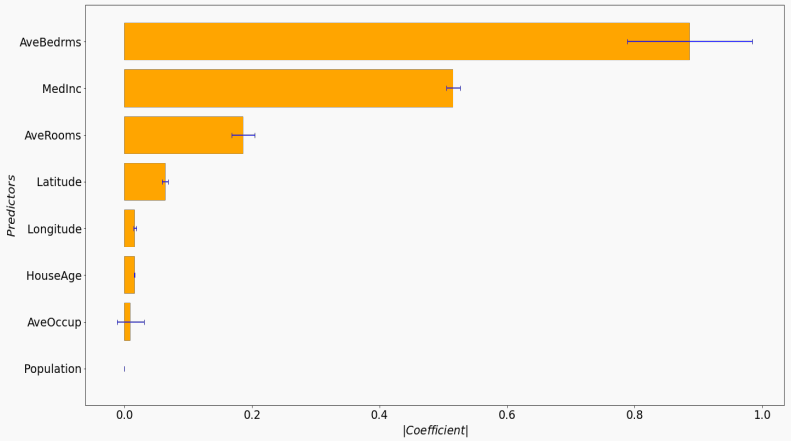
$\mu_{\hat{\beta}_i}$ is the mean of the bootstrap estimates of $\beta's$.
$\mathrm{SE}_{\hat{\beta}_i}$ is estimated by the standard deviation of the bootstrap coefficient estimates, $\sigma_{\hat{\beta}_i}$



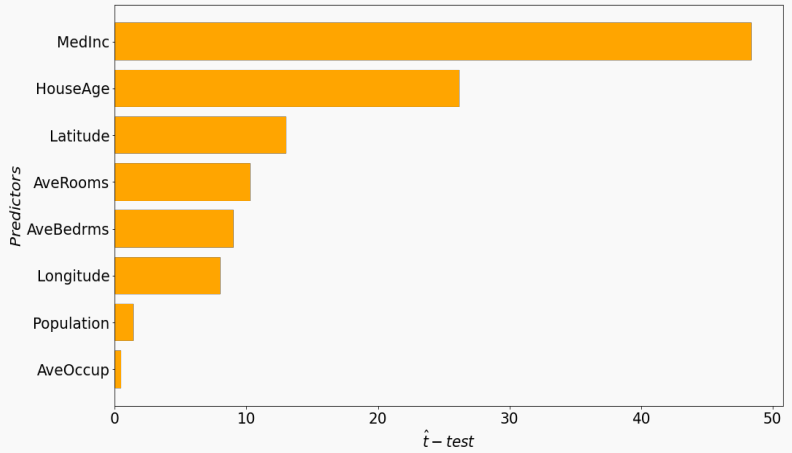
Feature Importance



The absolute value of the coefficients.



The absolute value of the coefficients over multiple bootstraps and includes the uncertainty of the coefficients.



The $\hat{t}$. Notice the rank of the importance has changed.

Feature Importance

Being ranked as the most important predictor does not necessarily mean that the predictor is related to the outcome.

How can we tell whether the apparent relationship is more than chance?

As with R^2 score, we compare the observed statistic ($\hat{t}$) with values calculated from random datasets in which the predictors and outcome are unrelated.



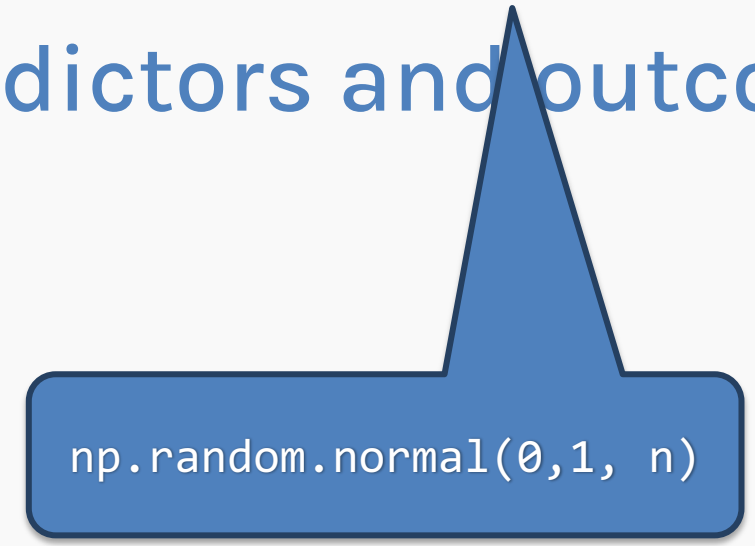
Feature Importance

What do we mean by random data?

We compare the observed statistic ($\hat{t}$) with values calculated from random datasets in which the predictors and outcome are unrelated.

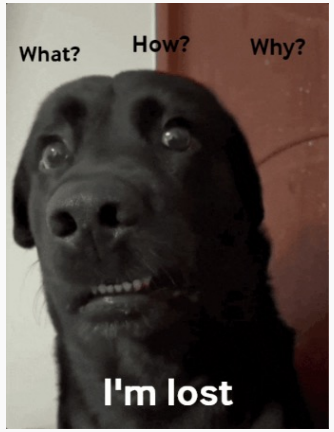
Feature Importance

We compare the observed statistic ($\hat{t}$) with values calculated from random datasets in which the predictors and outcome are unrelated.



Feature Importance

1. Generate one random dataset with the same number of observations, in which the predictors and outcome are unrelated.
2. Bootstrap this dataset. Fit the same regression model to each bootstrap resample and save the coefficients.
3. Calculate $\hat{t}_random$ for each predictor: divide the mean of its bootstrap coefficient estimates by their standard deviation, exactly as for the observed data: $\hat{t}_{random} = \hat{t}^R = \frac{\mu_{\hat{\beta}_j}}{\widehat{SE}_{\hat{\beta}_j}}$.
4. Repeat steps 1–3 for many random datasets. Plot a histogram of the resulting $\hat{t}_random$ values for each predictor.

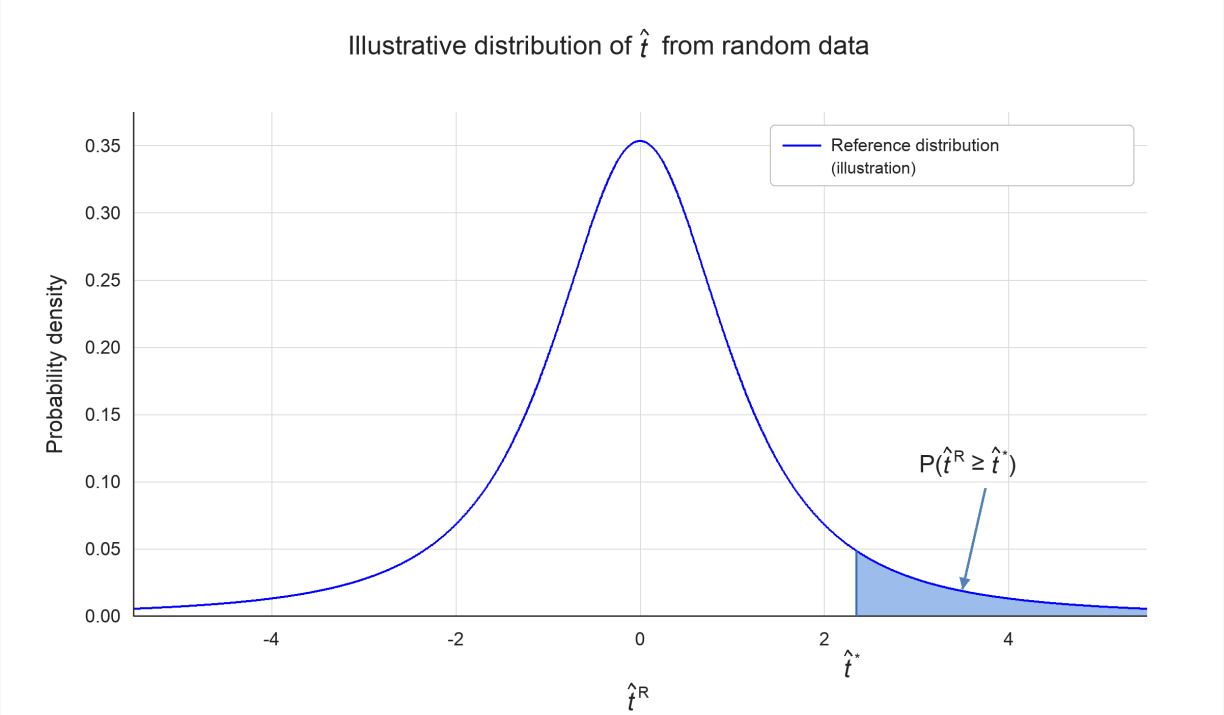


P-value

We compare the observed statistic from our model, $\hat{t}^*$, with $\hat{t}^R$ values calculated from random datasets in which the predictors and outcome are unrelated.

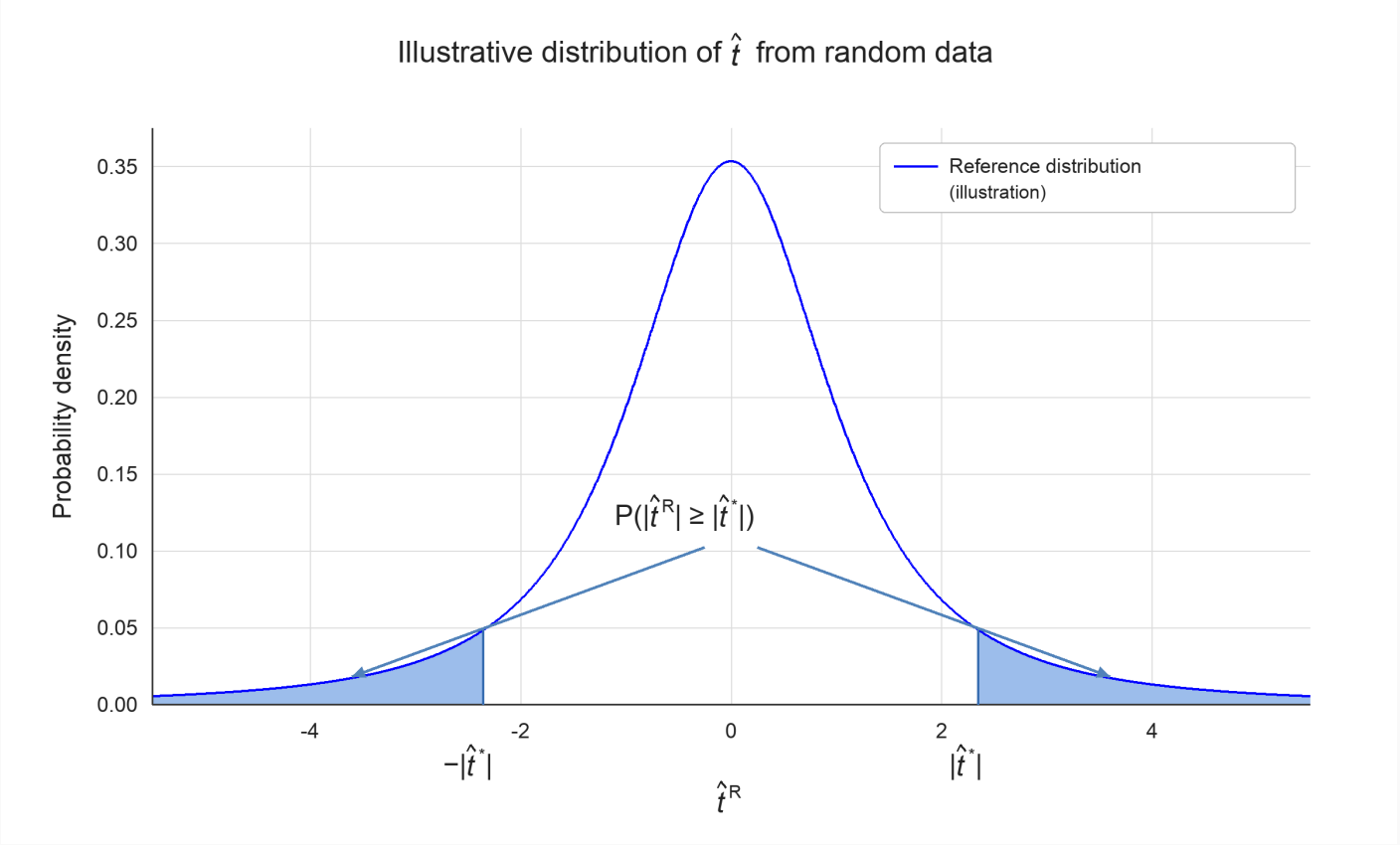
For a positive observed statistic, we first estimate the probability of obtaining a random-data statistic at least as large as the observed statistic:

$$P(\hat{t}^R \geq \hat{t}^\star).$$



P-value

Actually, we need to compare $|\hat{t}^{\star}|$, with the one calculated using random data, $|\hat{t}^{R}|$, we estimate the probability of observing $|\hat{t}^{R}| \geq |\hat{t}^{\star}|$.

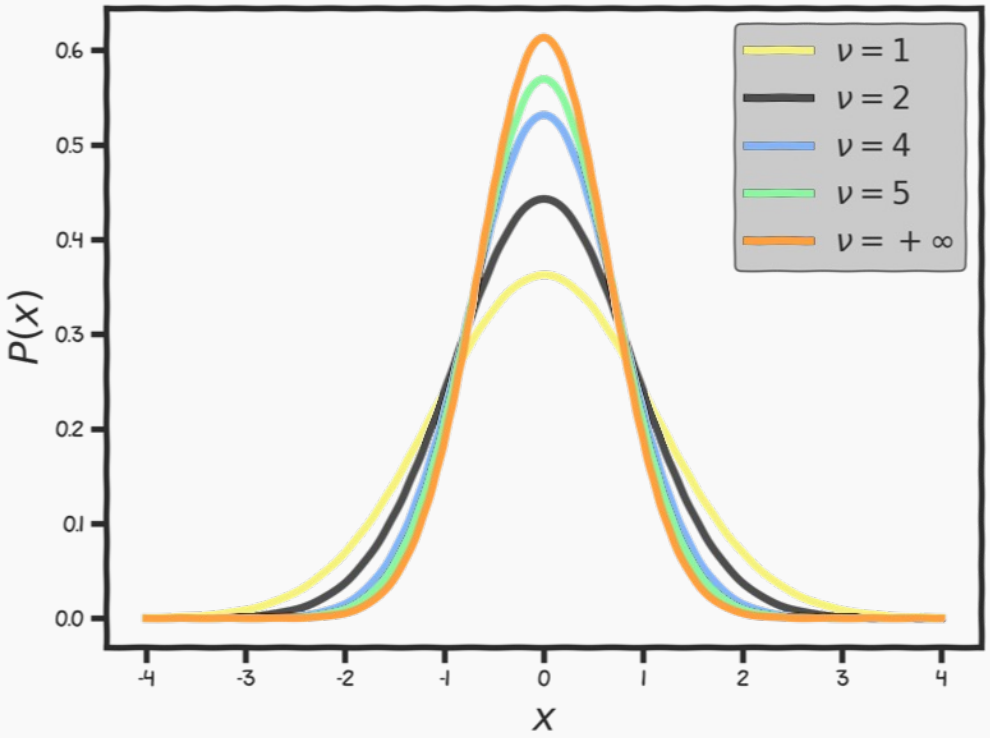


Permutation tests: another way to generate comparison data

- Previously, we generated random datasets in which the predictors and outcome were unrelated.
- Generating data from scratch requires assumptions about their distributions and the correlations among predictors. If these do not match the relevant no-association setting, the resulting p-values may be misleading.
- **Permutation uses the observed data instead.** Keep the predictors fixed and randomly shuffle the outcome across observations. This preserves the observed values and predictor correlations while breaking the predictor–outcome pairing.
- For each shuffled dataset, calculate the same $\hat{t}$ statistic using the bootstrap procedure.
- Repeat many times. Estimate the **two-sided p-value** as the fraction of shuffled-data statistics at least as far from zero as the observed $\hat{t}$.

Feature Importance

It turns out we do not have to do this: when the errors are roughly normal, the null distribution of the classical OLS t-statistic is approximately a known one, the Student-t distribution.



Student-t distribution, where ν is the degrees of freedom (number of data points minus number of predictors) = $n - (p + 1)$).

More about student-t and degrees of freedom

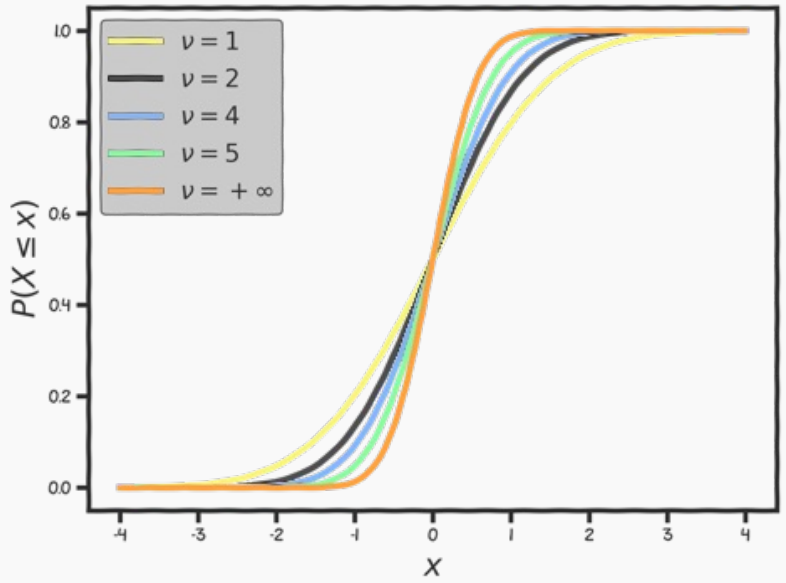
P-value

We call this probability the p-value:

$$p - value = P(|\hat{t}^R| \geq |\hat{t}^\star|)$$

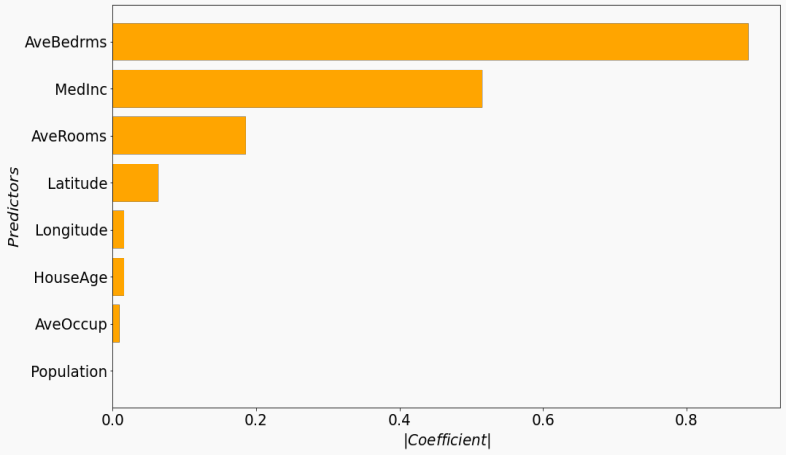
Small p-value indicates that it is unlikely to observe such a substantial association between the predictor and the response due to chance.
It is common to use p-value<0.05 as the threshold for significance.

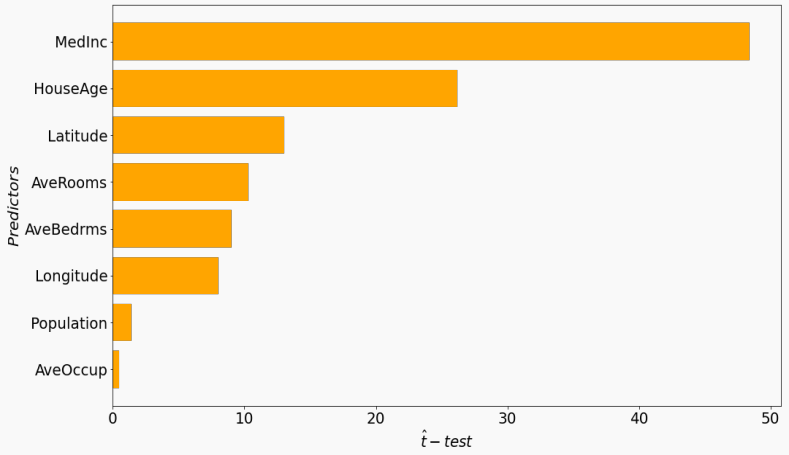
To calculate the p-value we use the cumulative distribution function (CDF) of the student-t. `stats model` a python library has a build-in function `stats.t.cdf()` which can be used to calculate this.

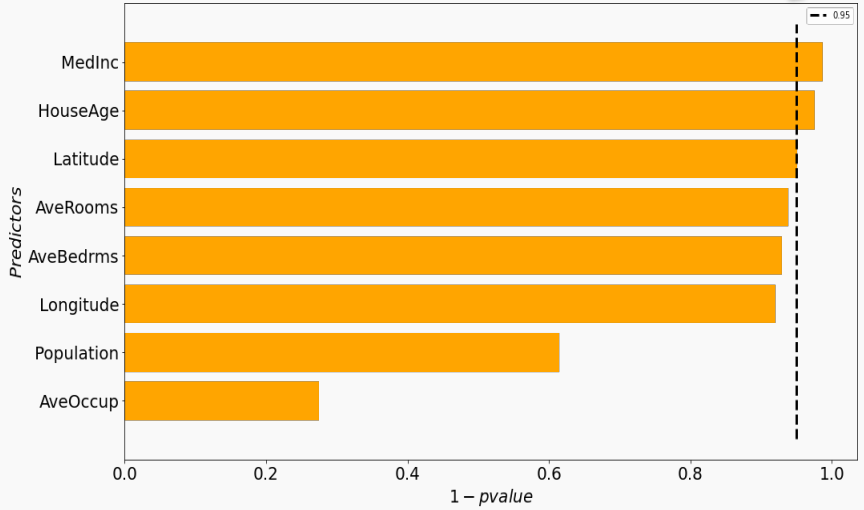


Feature Importance

Any predictor with a $1 - p$ value higher than this is considered important.







The absolute value of the coefficients over multiple bootstraps and includes the coefficients' uncertainty.

The $\hat{t}$. Notice the rank of the importance has changed.

Using the the p-value we also have which predictors are important. Note here we use 1-p.

Hypothesis Testing

Hypothesis testing is a formal process through which we evaluate the validity of a statistical hypothesis by considering evidence for or against the hypothesis gathered by random sampling of the data.

1. State the hypotheses, typically a **null hypothesis**, H_0 and an **alternative hypothesis**, H_1, that is the negation of the former.
2. Choose a type of analysis, i.e. how to use sample data to evaluate the null hypothesis. Typically, this involves choosing a single test statistic.
3. **Sample** data and compute the test statistic.
4. Use the value of the test statistic to either reject or not reject the null hypothesis.

Hypothesis Testing

1. State Hypothesis:

Null hypothesis:

H_0: There is no relation between X_j and Y , $(\beta_j = 0)$.

The alternative:

H_a: There is some relation between X_j and Y, $(\beta_j \neq 0)$.

2. **Choose test statistics**

$$\hat{t} = \frac{\mu_{\hat{\beta}_1}}{\widehat{SE}_{\hat{\beta}_1}}$$

Hypothesis Testing

3. Sample:

Using bootstrap we can estimate $\hat{\beta}_1's$, and $\mu_{\hat{\beta}_1}$ and $\widehat{SE}_{\hat{\beta}_1}$ and the $\hat{t}$.

4. **Reject or not reject the hypothesis:**

We compute ***p-value*** **,** the probability of observing any value equal to $|\hat{t}|$ or larger, from random data.

If p-value < p-value-threshold we **reject the null**.

Not Done Yet

